Modal Logic: Tableaux Examples

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Example 2:

$(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ in K

Example 2: $(\Diamond A \vee \Diamond B) \to \Diamond(A \vee B)$ in K

Let's show this is a theorem in K.

1.	$\mathbb{F}\ 1, (\Diamond A \vee \Diamond B) \to \Diamond(A \vee B)$ ✔	Assumption
2.	$\mathbb{T}\ 1, \Diamond A \vee \Diamond B$	$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \vee B)$	$\to\mathbb{F}\ 1$

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ ✔	Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ ✔	$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$	$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A \qquad \mathbb{T}\ 1, \Diamond B$	$\lor\mathbb{T}\ 2$

The disjunction branches. We can't do anything with the other line yet because there isn't yet a 1.1.

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ ✔		Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ ✔		$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$		$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A$ ✔	$\mathbb{T}\ 1, \Diamond B$	$\lor\mathbb{T}\ 2$
5.	$\mathbb{T}\ 1.1, A$		$\Diamond\mathbb{T}\ 4$

On the left branch, the True $\Diamond$ rule introduces world 1.1 with A true there.

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ $\checkmark$		Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ $\checkmark$		$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$		$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A$ $\checkmark$	$\mathbb{T}\ 1, \Diamond B$	$\lor\mathbb{T}\ 2$
5.	$\mathbb{T}\ 1.1, A$		$\Diamond\mathbb{T}\ 4$
6.	$\mathbb{F}\ 1.1, A \lor B$		$\Diamond\mathbb{F}\ 3$

Now world 1.1 exists, so the False $\Diamond$ rule (line 3) applies: $A \lor B$ must be false at 1.1.

1.	$\mathbb{F}\, 1, (\Diamond A \vee \Diamond B) \to \Diamond(A \vee B)$ ✔	Assumption
2.	$\mathbb{T}\, 1, \Diamond A \vee \Diamond B$ ✔	$\to\mathbb{F}\, 1$
3.	$\mathbb{F}\, 1, \Diamond(A \vee B)$	$\to\mathbb{F}\, 1$
4.	$\mathbb{T}\, 1, \Diamond A$ ✔ $\quad$ $\mathbb{T}\, 1, \Diamond B$	$\vee\mathbb{T}\, 2$
5.	$\mathbb{T}\, 1.1, A$	$\Diamond\mathbb{T}\, 4$
6.	$\mathbb{F}\, 1.1, A \vee B$ ✔	$\Diamond\mathbb{F}\, 3$
7.	$\mathbb{F}\, 1.1, A$	$\vee\mathbb{F}\, 6$
	$\otimes$	

The False $\vee$ rule gives $\mathbb{F}\, 1.1{:}A$, which clashes with line 5. The left branch closes. Now work on the right branch.

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ ✔		Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ ✔		$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$		$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A$ ✔	$\mathbb{T}\ 1, \Diamond B$ ✔	$\lor\mathbb{T}\ 2$
5.	$\mathbb{T}\ 1.1, A$		$\Diamond\mathbb{T}\ 4$
6.	$\mathbb{F}\ 1.1, A \lor B$ ✔		$\Diamond\mathbb{F}\ 3$
7.	$\mathbb{F}\ 1.1, A$		$\lor\mathbb{F}\ 6$
8.	$\otimes$	$\mathbb{T}\ 1.2, B$	$\Diamond\mathbb{T}\ 4$

On the right branch, the True $\Diamond$ rule introduces world 1.2 with B true there.

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ ✔		Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ ✔		$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$		$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A$ ✔	$\mathbb{T}\ 1, \Diamond B$ ✔	$\lor\mathbb{T}\ 2$
5.	$\mathbb{T}\ 1.1, A$		$\Diamond\mathbb{T}\ 4$
6.	$\mathbb{F}\ 1.1, A \lor B$ ✔		$\Diamond\mathbb{F}\ 3$
7.	$\mathbb{F}\ 1.1, A$		$\lor\mathbb{F}\ 6$
8.	$\otimes$	$\mathbb{T}\ 1.2, B$	$\Diamond\mathbb{T}\ 4$
9.		$\mathbb{F}\ 1.2, A \lor B$	$\Diamond\mathbb{F}\ 3$

World 1.2 now exists, so the False $\Diamond$ rule (line 3) applies again: $A \lor B$ must be false at 1.2.

1.	$\mathbb{F}\ 1, (\Diamond A \lor \Diamond B) \to \Diamond(A \lor B)$ ✔		Assumption
2.	$\mathbb{T}\ 1, \Diamond A \lor \Diamond B$ ✔		$\to\mathbb{F}\ 1$
3.	$\mathbb{F}\ 1, \Diamond(A \lor B)$		$\to\mathbb{F}\ 1$
4.	$\mathbb{T}\ 1, \Diamond A$ ✔	$\mathbb{T}\ 1, \Diamond B$ ✔	$\lor\mathbb{T}\ 2$
5.	$\mathbb{T}\ 1.1, A$		$\Diamond\mathbb{T}\ 4$
6.	$\mathbb{F}\ 1.1, A \lor B$ ✔		$\Diamond\mathbb{F}\ 3$
7.	$\mathbb{F}\ 1.1, A$		$\lor\mathbb{F}\ 6$
8.	$\otimes$	$\mathbb{T}\ 1.2, B$	$\Diamond\mathbb{T}\ 4$
9.		$\mathbb{F}\ 1.2, A \lor B$ ✔	$\Diamond\mathbb{F}\ 3$
10.		$\mathbb{F}\ 1.2, A$	$\lor\mathbb{F}\ 9$
11.		$\mathbb{F}\ 1.2, B$	$\lor\mathbb{F}\ 9$
		$\otimes$	

Both branches close, so this **is** valid in K.